



Agnel Charities

Fr. C. Rodrigues Institute of Technology, Vashi, Navi-Mumbai

An Autonomous Institute Permanently Affiliated to University of Mumbai

Department of Computer Engineering

Project Proposal

Group No	1	
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Project Category-I	Outhouse	
Project Category-II	Application	
Mapped SDG	3: Good Health and Well-Being; 9: Industry, Innovation, and Infrastructure; 10: Reduced Inequalities	
Project Proposal	Domain: AI-ML, Human -Computer Interaction (HCI)	
	Title: Thought – to – Speech Converter	
	Project Idea: Develop a Brain–Computer Interface (BCI) system using a DIY neuroscience kit to acquire brain signals (EEG/biopotential signals), process them using AI techniques, and convert recognized mental patterns into words, sentences, and synthesized speech. The project explores the integration of neuroscience, signal processing, machine learning, and natural language generation to create an assistive communication system. Objectives: • Study the fundamentals of Brain–Computer Interfaces (BCIs) and EEG signal acquisition. • Collect and preprocess brain signals using a DIY neuroscience kit. • Extract meaningful signal features associated with predefined mental states or commands. • Develop machine learning models to classify thoughts, intentions, or cognitive patterns.	

- Construct a Thought-to-Word-to-Sentence-to-Speech (TWSS) pipeline that generates meaningful textual and audio output.
- Evaluate the effectiveness of the system as an assistive communication tool.

Feasibility	Study:
Technical Feasibility: The project uses an accessible, well-documented hardware platform (Upside Down Labs' DIY neuroscience kit) that provides EEG/EMG/EOG signal acquisition without requiring clinical-grade equipment. Signal processing and ML pipelines can be built with mature, well-supported open-source libraries (MNE-Python, NeuroKit2, Scikit-learn, TensorFlow/PyTorch), and the team can validate its approach against established public datasets (PhysioNet, BCI Competition) before relying on self-collected data, which reduces technical risk. Combining a classification engine with existing TTS libraries (pytt3x3, Coqui TTS, gTTS) is a proven integration pattern, so the full Thought-to-Word-to-Sentence-to-Speech pipeline is achievable within the team's skill set. Operational Feasibility: Assistive communication tools for individuals with motor impairments address a well-recognized real-world need, and a low-cost DIY-kit-based approach makes the system more accessible than commercial BCI devices. Starting with a small, fixed vocabulary of predefined commands ("Yes," "No," "Need Water," etc.) keeps the initial classification task operationally realistic, with room to scale complexity later. Economic Feasibility: The hardware kit and all proposed software are low-cost or free/open-source, and public EEG datasets eliminate the need for expensive proprietary data. This keeps overall project cost minimal and appropriate for an academic setting. Data Feasibility: Multiple public EEG datasets (PhysioNet, BCI Competition, OpenMIIR, Motor Imagery) provide a solid foundation for model development and benchmarking. Team-generated recordings and simulated signals can supplement this for early-stage testing and calibration, though the team should budget extra time for signal quality issues common in low-cost EEG hardware (noise, electrode placement, motion artifacts). Schedule Feasibility: The project's modular structure (signal acquisition → preprocessing → feature extraction → classification → sentence generation → speech synthesis) allows incremental, parallel development and testing, making it achievable within a semester timeline. The most time-sensitive/uncertain stage is signal classification	

	accuracy, so this should be prioritized early with a fallback of a smaller command set if classification proves harder than expected. Ethical & Safety Feasibility: Since the system involves biosignal acquisition from human subjects, the team should ensure informed consent from all participants (including team members) during data collection, follow basic electrical safety practices with the DIY kit, and be transparent about the system's current accuracy limitations if demonstrated as an assistive tool (i.e., not positioning it as a validated medical device).
	Technological Stack: Programming Language: Python Signal Acquisition & Processing: Standard Python scientific computing libraries (e.g., NumPy, SciPy) along with an EEG-specific processing library for filtering, artifact removal, and feature extraction Machine Learning: A general-purpose ML library (e.g., Scikit-learn) for initial classification models, with the option to move to a deep learning framework (TensorFlow/PyTorch) later if simpler models prove insufficient Natural Language Generation: A lightweight rule-based or template-based sentence construction approach initially, with room to explore more advanced NLP tools if the project scope allows Text-to-Speech: Any accessible, easy-to-integrate TTS library or API for converting generated sentences to audio Data Visualization: A standard plotting library for visualizing raw and processed signals User Interface: A simple web-based interface framework for demonstrating the pipeline (e.g., Streamlit for quick prototyping) Development Tools: Jupyter Notebook for experimentation, VS Code for development, Git/GitHub for version control

We approve the project as described above, and authorize the team to proceed.

Project Guide

Dr. Rakhi Kalantri

(Project Coordinator)